Flutter Lab External Exam - Optimized Codes

This document contains **optimized Flutter/Dart codes for each week** as per your lab manual. Output remains exactly the same, but code is cleaner, shorter, and exam-friendly.

✅WEEK 1 - Introduction

(No coding required – Theory & environment overview)

✅WEEK 1 - Installation of Android Studio & Flutter

(No program – Setup process only)

✅WEEK 2 - Hello World Application

Optimized Flutter Code:

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text('Hello App')),
        body: Center(child: Text('Hello World')),
      ),
    );
  }
}
```

✅WEEK 3 - Decision Making & Loops (Dart)

Optimized Dart Code:

```
void main() {
  int n = 5;
```

```
  if (n % 2 == 0) {
    print('Even Number');
  } else {
    print('Odd Number');
  }

  for (int i = 1; i <= 5; i++) {
    print('Value: $i');
  }
}
```

✅WEEK 4 - User Defined Functions

```
void main() {
  int result = add(10, 20);
  print('Sum: $result');
}

int add(int a, int b) => a + b;
```

✅WEEK 5 - Object Oriented Programming

```
class Student {
  String name;
  int age;

  Student(this.name, this.age);

  void display() {
    print('Name: $name, Age: $age');
  }
}

void main() {
  Student s = Student('Akshay', 21);
  s.display();
}
```

✅WEEK 6 - Basic Widgets (Text, Image, Icon)

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: const Text('Basic Widgets')),
        body: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: const [
            Text('Flutter Widgets', style: TextStyle(fontSize: 20)),
            Icon(Icons.star, size: 40),
          ],
        ),
      ),
    );
  }
}
```

✅WEEK 7 - Layout Widgets

A) Single Child Widget (Container inside Center)

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: const Text('Single Child Widget')),
        body: Center(
          child: Container(
            height: 100,
            width: 100,
            color: Colors.blue,
```

```
            child: const Center(
              child: Text('Single'),
            ),
          ),
        ),
      ),
    );
  }
}
```

B) Multiple Child Widget (Row & Column)

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: const Text('Multiple Child Widgets')),
        body: Column(
          children: [
            Container(height: 60, color: Colors.red),
            Row(
              children: [
                Expanded(child: Container(height: 60, color: Colors.green)),
                Expanded(child: Container(height: 60, color: Colors.orange)),
              ],
            ),
          ],
        ),
      ),
    );
  }
}
```

✅WEEK 8 - Gesture Detector

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatefulWidget {
```

```
  const MyApp({super.key});

  @override
  State<MyApp> createState() => _MyAppState();
}

class _MyAppState extends State<MyApp> {
  String msg = 'Tap the box';

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: const Text('Gesture Demo')),
        body: Center(
          child: GestureDetector(
            onTap: () {
              setState(() => msg = 'Box Tapped');
            },
            child: Container(
              padding: const EdgeInsets.all(30),
              color: Colors.amber,
              child: Text(msg),
            ),
          ),
        ),
      ),
    );
  }
}
```

✅WEEK 9 & 10 - Registration Form

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: const Text('Registration Form')),
        body: Padding(
          padding: const EdgeInsets.all(16),
          child: Column(
```

```
            children: const [
              TextField(decoration: InputDecoration(labelText: 'Name')),
              TextField(decoration: InputDecoration(labelText: 'Email')),
              TextField(decoration: InputDecoration(labelText: 'Password'),
obscureText: true),
              SizedBox(height: 20),
              ElevatedButton(onPressed: null, child: Text('Register')),
            ],
          ),
        ),
      ),
    );
  }
}
```

✅WEEK 11 - Flutter Calendar

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: const Text('Calendar')),
        body: Center(child: Text(DateTime.now().toString())),
      ),
    );
  }
}
```

✅WEEK 12 - Animated Text

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatefulWidget {
  const MyApp({super.key});

  @override
```

```
  State<MyApp> createState() => _MyAppState();
}

class _MyAppState extends State<MyApp> with SingleTickerProviderStateMixin {
  late AnimationController controller;

  @override
  void initState() {
    super.initState();
    controller = AnimationController(vsync: this, duration: const
Duration(seconds: 2))..repeat();
  }

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: const Text('Animated Text')),
        body: Center(
          child: FadeTransition(
            opacity: controller,
            child: const Text('Flutter Animation', style:
TextStyle(fontSize: 24)),
          ),
        ),
      ),
    );
  }
}
```

✅All programs are optimized and ready for lab external.

If you want: - PDF version - Viva explanation - Output screenshots - Comments added for writing

Just tell me ♀️